

Questions: Inventive Cognition

Analytical Questions

1. Define analogical reasoning in Active Inference terms. How does Gentner's structure-mapping theory relate to the concept of shared generative model structure across domains?
2. Gutenberg's printing press analogy from the wine press transferred relational structure (uniform pressure across a surface) while discarding surface features (grapes vs. ink). Explain why the success of an analogy depends on correctly identifying which structural relationships are preserved and which are not.
3. What distinguishes productive combinatorial creativity from random concatenation? Use Boden's criteria (complementarity, emergence, coherence) to explain why the smartphone was a productive combination while other multi-function devices have failed.
4. Explain TRIZ contradictions in Active Inference terms. Why does the framework treat contradictions as resolvable through model transformation rather than as inherent trade-offs that must be accepted?
5. Mental simulation in Active Inference corresponds to counterfactual policy evaluation. How does the accuracy of the generative model determine the value of mental simulation? What happens when an inventor simulates with an inaccurate model?
6. Compare the cognitive demands of analogical reasoning, combinatorial creativity, and TRIZ. Which cognitive strategy requires the broadest generative model? Which requires the deepest? Which is most systematic?
7. The concept of cognitive fluency — generating many alternatives before converging — seems inefficient. Use the Active Inference framework to explain why maintaining high entropy in the policy distribution during early ideation actually reduces overall free energy in the long run.

8. Altshuller developed TRIZ by analyzing 200,000 patents and identifying recurring solution patterns. How does this empirical approach to building a library of model-transformation operations relate to the Active Inference concept of learning generative model structure from data?
9. The "horseless carriage" analogy led to early automobile designs that resembled horse-drawn carriages. Analyze this as a false analogy — which structural relationships appeared to transfer but actually did not? What cognitive process could have helped early automobile designers detect the analogy's failure earlier?
10. Tesla's practice of mental simulation (building and testing machines in imagination) and Edison's practice of physical prototyping (building thousands of variants in the laboratory) represent opposite ends of a cognitive spectrum. Using Active Inference, explain the trade-offs between each approach and why the optimal strategy typically involves alternation between them.

Applied Questions

1. State your invention problem in one sentence. Then abstract it to its structural essence by removing domain-specific details. Can you express it as a pattern of relationships, flows, or transformations?
2. Identify three source domains where a problem structurally similar to yours has been solved. For each, describe the analogy and assess how well the structural relationships transfer to your domain.
3. List eight existing technologies or methods relevant to your invention domain. Create three novel combinations and assess each for complementarity, emergence, and coherence. Which combination is most promising?
4. Identify a core contradiction in your invention problem (improving parameter A worsens parameter B). Apply three different TRIZ principles to resolve it. Which resolution produces the most viable concept?

5. Select your most promising invention concept and run a detailed mental simulation.
Walk through a complete use scenario step by step, noting potential failure modes, edge cases, and surprising consequences. What did the simulation reveal?
6. Apply the SCAMPER method to your current best invention concept: Substitute one component, Combine it with something, Adapt it for a new context, Modify its scale, Put it to another use, Eliminate a component, and Reverse an assumption. Which transformation produces the most interesting variant?
7. Compare the invention concepts you generated using analogy, combination, and TRIZ. Do they converge on similar solutions or diverge into different regions of the solution space? What does this tell you about your problem?
8. Identify a false analogy you have been using in your invention thinking — a structural mapping that seems right but actually breaks down under scrutiny. Where does the analogy fail, and what alternative framing might work better?
9. Design a "cognitive protocol" for your daily inventive practice — a sequence of cognitive strategies you would apply when working on new problems. Specify when you would use each strategy, how long you would spend on it, and what output you expect from each step.
10. Choose a famous invention and reverse-engineer the cognitive process that produced it. Which cognitive strategy (analogy, combination, TRIZ, mental simulation) was most important? What generative models did the inventor need to hold simultaneously?